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MCY 612  
Lab 2 VPC  
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## MCY 612 Cloud Security Lab#2: Virtual Private Cloud (VPC)

In this lab, you will study the Virtual Private Cloud (VPC). You will use the AWS CLI to build a VPC.

### Tools and Materials

AWS credentials  
AWS CloudShell CLI  
Ubuntu Server 20.04 LTS AMI: ami-09e67e426f25ce0d7

**Note: You need to replace words in *italics* with your information.**

### Part 1 Create a VPC and a subnet

1. Start your AWS CloudShell terminal. Create a VPC via the following command:

```
aws ec2 create-vpc --cidr-block 10.6.8.0/26
```

**What is the complete output? What is your VpcId?**

```
~ $ aws ec2 create-vpc --cidr-block 10.6.8.0/26
{
  "Vpc": {
    "OwnerId": "881490118823",
    "InstanceTenancy": "default",
    "Ipv6CidrBlockAssociationSet": [],
    "CidrBlockAssociationSet": [
      {
        "AssociationId": "vpc-cidr-assoc-0075ec512b32cb75d",
        "CidrBlock": "10.6.8.0/26",
        "CidrBlockState": {
          "State": "associated"
        }
      }
    ],
    "IsDefault": false,
    "VpcId": "vpc-0cb3e30db617f6175",
    "State": "pending",
    "CidrBlock": "10.6.8.0/26",
    "DhcpOptionsId": "dopt-08b45b45933d7afb0"
  }
}
```

2. Create a subnet in the VPC created at step 1 via the following command:

```
aws ec2 create-subnet --vpc-id your_VpcId --cidr-block 10.6.8.0/28 --availability-zone us-east-1c
```

**What is your command and its complete output? What is your SubnetId? How many IP addresses are there in your subnet?**

```

~ $ aws ec2 create-subnet --vpc-id vpc-0cb3e30db617f6175 --cidr-block 10.6.8.0/28 --availability-zone us-east-1c
{
  "Subnet": {
    "AvailabilityZoneId": "use1-az2",
    "OwnerId": "881490118823",
    "AssignIpv6AddressOnCreation": false,
    "Ipv6CidrBlockAssociationSet": [],
    "SubnetArn": "arn:aws:ec2:us-east-1:881490118823:subnet/subnet-09df9a6645aea6ede",
    "EnableDns64": false,
    "Ipv6Native": false,
    "PrivateDnsNameOptionsOnLaunch": {
      "HostnameType": "ip-name",
      "EnableResourceNameDnsARecord": false,
      "EnableResourceNameDnsAAAARecord": false
    },
    "SubnetId": "subnet-09df9a6645aea6ede",
    "State": "available",
    "VpcId": "vpc-0cb3e30db617f6175",
    "CidrBlock": "10.6.8.0/28",
    "AvailableIpAddressCount": 11,
    "AvailabilityZone": "us-east-1c",
    "DefaultForAz": false,
    "MapPublicIpOnLaunch": false
  }
}

```

3. Create an internet gateway for your VPC with the following command:

```
aws ec2 create-internet-gateway
```

**What is your internet gateway? What is the complete output? What is your InternetGatewayId?**

```

~ $ aws ec2 create-internet-gateway
{
  "InternetGateway": {
    "Attachments": [],
    "InternetGatewayId": "igw-0cc59d6a34d65d113",
    "OwnerId": "881490118823",
    "Tags": []
  }
}
~ $

```

4. Attach the internet gateway to your VPC to enable connectivity between the internet and your VPC with the following command:

```
aws ec2 attach-internet-gateway --vpc-id your_VpcId --internet-gateway-id your_InternetGatewayId
```

**What is your command?**

```
~ $ aws ec2 attach-internet-gateway --vpc-id vpc-0cb3e30db617f6175 --internet-gateway-id igw-0cc59d6a34d65d113
```

5. Create a route table for your VPC with the following command:

```
aws ec2 create-route-table --vpc-id your_VpcId
```

**What is your command and its complete output? What is your RouteTableId?**

```

~ $ aws ec2 create-route-table --vpc-id vpc-0cb3e30db617f6175
{
  "RouteTable": {
    "Associations": [],
    "PropagatingVgws": [],
    "RouteTableId": "rtb-0940a3306905b7239",
    "Routes": [
      {
        "DestinationCidrBlock": "10.6.8.0/26",
        "GatewayId": "local",
        "Origin": "CreateRouteTable",
        "State": "active"
      }
    ],
    "Tags": [],
    "VpcId": "vpc-0cb3e30db617f6175",
    "OwnerId": "881490118823"
  },
  "ClientToken": "b49e4e04-1b7f-4d8d-9870-08f4028835b7"
}

```

6. Create a networking route in your route table within your VPC with the following command:

`aws ec2 create-route --route-table-id your_RouteTableId --destination-cidr-block 0.0.0.0/0 --gateway-id your_InternetGatewayId`

**What is your command and its complete output?**

```

~ $ aws ec2 create-route --route-table-id rtb-0940a3306905b7239 --destination-cidr-block 0.0.0.0/0 --gateway-id igw-0cc59d6a34d65d113
{
  "Return": true
}

```

7. Verify your route has been created and is active with the following command:

`aws ec2 describe-route-tables --route-table-id your_RouteTableId`

**What is your command and its complete output?**

```

~ $ aws ec2 describe-route-tables --route-table-id rtb-0940a3306905b7239
{
  "RouteTables": [
    {
      "Associations": [],
      "PropagatingVgws": [],
      "RouteTableId": "rtb-0940a3306905b7239",
      "Routes": [
        {
          "DestinationCidrBlock": "10.6.8.0/26",
          "GatewayId": "local",
          "Origin": "CreateRouteTable",
          "State": "active"
        },
        {
          "DestinationCidrBlock": "0.0.0.0/0",
          "GatewayId": "igw-0cc59d6a34d65d113",
          "Origin": "CreateRoute",
          "State": "active"
        }
      ],
      "Tags": [],
      "VpcId": "vpc-0cb3e30db617f6175",
      "OwnerId": "881490118823"
    }
  ]
}

```

8. Associate the subnet with the route table in your VPC with the following command. This association causes traffic from your subnet to be routed according to the routes in your route table.

`aws ec2 associate-route-table --subnet-id your_SubnetId --route-table-id your_RouteTableId`

### What are your command and its complete output?

```
~ $ aws ec2 associate-route-table --subnet-id subnet-09df9a6645aea6ede --route-table-id rtb-0940a3306905b7239
{
  "AssociationId": "rtbassoc-0ca3a6f2140910dbe",
  "AssociationState": {
    "State": "associated"
  }
}
```

9. Modify the public IP addressing attribute with the following command so that the EC2 Instance launched into your subnet automatically receives a public IP address.

```
aws ec2 modify-subnet-attribute --subnet-id you_SubnetId --map-public-ip-on-launch
```

### What is your command?

```
~ $ aws ec2 modify-subnet-attribute --subnet-id subnet-09df9a6645aea6ede --map-public-ip-on-launch
```

Do the following steps to test if the VPC, subnet, routing table you created at the previous steps function.

10. Create a security group via the following command:

```
aws ec2 create-security-group --group-name your_username --description "my security group" --vpc-id your_VpcId
```

### What are your command and its complete output? What is your security group Id?

```
~ $ aws ec2 create-security-group --group-name carrd12 --description "my security group" --vpc-id vpc-0cb3e30db617f6175
{
  "GroupId": "sg-0b1cfa43fd7ea602b",
  "SecurityGroupArn": "arn:aws:ec2:us-east-1:881490118823:security-group/sg-0b1cfa43fd7ea602b"
}
```

11. Add a rule, which allows incoming traffic over port 22 for SSH, to the security group with the following command:

```
aws ec2 authorize-security-group-ingress --group-id your_security_group_id --protocol tcp --port 22 --cidr 0.0.0.0/0
```

```
~ $ aws ec2 authorize-security-group-ingress --group-id sg-0b1cfa43fd7ea602b --protocol tcp --port 22 --cidr 0.0.0.0/0
{
  "Return": true,
  "SecurityGroupRules": [
    {
      "SecurityGroupRuleId": "sgr-034a3b837fe227642",
      "GroupId": "sg-0b1cfa43fd7ea602b",
      "GroupOwnerId": "881490118823",
      "IsEgress": false,
      "IpProtocol": "tcp",
      "FromPort": 22,
      "ToPort": 22,
      "CidrIpv4": "0.0.0.0/0",
      "SecurityGroupRuleArn": "arn:aws:ec2:us-east-1:881490118823:security-group-rule/sgr-034a3b837fe227642"
    }
  ]
}
```

12. If you did not delete your keypair that you created in MCY 611, you need to delete your keypair with the following command:

```
aws ec2 delete-key-pair --key-name your_username-key
```

```
~ $ aws ec2 delete-key-pair --key-name carrd12-key
{
  "Return": true,
  "KeyPairId": "key-0cd79cdd0f55e71e2"
}
```

13. Create a key pair with the following command:

```
aws ec2 create-key-pair --key-name your_username-key --query 'KeyMaterial' --output text > devenv-key.pem
```

```
~ $ aws ec2 create-key-pair --key-name carrd12-key --query 'KeyMaterial' --output text > devenv-key.pem
```

Modify permissions to the devenv-key.pem file with the following command so that only you can access it:

```
chmod 400 devenv-key.pem
```

14. Launch an EC2 instance into your VPC with the following command:

```
aws ec2 run-instances --image-id ami-09e67e426f25ce0d7 --count 1 --instance-type t3.micro --key-name your_username-key --security-group-ids your_security_group_id --subnet-id your_SubnetId
```

```
~ $ aws ec2 run-instances --image-id ami-09e67e426f25ce0d7 --count 1 --instance-type t3.micro --key-name carrd12-key --security-group-ids sg-0b1cfa43fd7ea602b --subnet-id subnet-09df9a6645aea6ede
{
  "ReservationId": "r-097dbfde1cf51977e",
  "OwnerId": "881490118823",
  "Groups": [],
  "Instances": [
    {
      "Architecture": "x86_64",
      "BlockDeviceMappings": [],
      "ClientToken": "e9dec8e2-e9ea-4365-a609-d518e3a54634",
      "EbsOptimized": false,
      "EnaSupport": true,
      "Hypervisor": "xen",
      "NetworkInterfaces": [
        {
          "Attachment": {
            "AttachTime": "2025-03-29T01:31:40+00:00",
            "AttachmentId": "eni-attach-0ea6d284323a0df35",
            "DeleteOnTermination": true,
            "DeviceIndex": 0,
            "Status": "attaching",
```

```
            "Status": "attaching",
            "NetworkCardIndex": 0
          },
          "Description": "",
          "Groups": [
            {
              "GroupId": "sg-0b1cfa43fd7ea602b",
              "GroupName": "carrd12"
            }
          ],
          "Ipv6Addresses": [],
          "MacAddress": "12:a6:ad:3b:d1:bf",
          "NetworkInterfaceId": "eni-030f0b1eac10daf70",
          "OwnerId": "881490118823",
          "PrivateIpAddress": "10.6.8.13",
          "PrivateIpAddresses": [
            {
              "Primary": true,
              "PrivateIpAddress": "10.6.8.13"
            }
          ],
          "SourceDestCheck": true,
          "Status": "pending",
          "SubnetId": "subnet-09df9a6645aea6ede",
          "Tags": [],
          "UserData": ""
        }
      ]
    }
  ]
}
```

```

        "SourceDestCheck": true,
        "Status": "in-use",
        "SubnetId": "subnet-09df9a6645aea6ede",
        "VpcId": "vpc-0cb3e30db617f6175",
        "InterfaceType": "interface",
        "Operator": {
            "Managed": false
        }
    },
    "RootDeviceName": "/dev/sda1",
    "RootDeviceType": "ebs",
    "SecurityGroups": [
        {
            "GroupId": "sg-0b1cfa43fd7ea602b",
            "GroupName": "carrd12"
        }
    ],
    "SourceDestCheck": true,
    "StateReason": {
        "Code": "pending",
        "Message": "pending"
    }
}

```

```

        "Message": "pending"
    },
    "VirtualizationType": "hvm",
    "CpuOptions": {
        "CoreCount": 1,
        "ThreadsPerCore": 2
    },
    "CapacityReservationSpecification": {
        "CapacityReservationPreference": "open"
    },
    "MetadataOptions": {
        "State": "pending",
        "HttpTokens": "optional",
        "HttpPutResponseHopLimit": 1,
        "HttpEndpoint": "enabled",
        "HttpProtocolIpv6": "disabled",
        "InstanceMetadataTags": "disabled"
    },
    "EnclaveOptions": {
        "Enabled": false
    },
    "PrivateDnsNameOptions": {
        "PrivateDnsName": "private-dns-name"
    }
}

```

```

    "PrivateDnsNameOptions": {
      "HostnameType": "ip-name",
      "EnableResourceNameDnsARecord": false,
      "EnableResourceNameDnsAAAARecord": false
    },
    "MaintenanceOptions": {
      "AutoRecovery": "default"
    },
    "CurrentInstanceBootMode": "legacy-bios",
    "Operator": {
      "Managed": false
    },
    "InstanceId": "i-05746087d8cc178ca",
    "ImageId": "ami-09e67e426f25ce0d7",
    "State": {
      "Code": 0,
      "Name": "pending"
    },
    "PrivateDnsName": "ip-10-6-8-13.ec2.internal",
    "PublicDnsName": "",
    "StateTransitionReason": "",
    "KeyName": "carrd12-key",
    "AmiLaunchIndex": 0,
    "ProductCodes": [],
    "InstanceType": "t3.micro",
    "LaunchTime": "2025-03-29T01:31:40+00:00",
    "Placement": {
      "GroupName": "",
      "Tenancy": "default",
      "AvailabilityZone": "us-east-1c"
    },
    "Monitoring": {
      "State": "disabled"
    },
    "SubnetId": "subnet-09df9a6645aea6ede",
    "VpcId": "vpc-0cb3e30db617f6175",
    "PrivateIpAddress": "10.6.8.13"
  }
}

```

```

    "PublicDnsName": "",
    "StateTransitionReason": "",
    "KeyName": "carrd12-key",
    "AmiLaunchIndex": 0,
    "ProductCodes": [],
    "InstanceType": "t3.micro",
    "LaunchTime": "2025-03-29T01:31:40+00:00",
    "Placement": {
      "GroupName": "",
      "Tenancy": "default",
      "AvailabilityZone": "us-east-1c"
    },
    "Monitoring": {
      "State": "disabled"
    },
    "SubnetId": "subnet-09df9a6645aea6ede",
    "VpcId": "vpc-0cb3e30db617f6175",
    "PrivateIpAddress": "10.6.8.13"
  }
}

```

```

]
}
~ $

```

15. Use the SSH command to log into your instance and then run "ping -c 5 google.com"

The purpose of the ping command is to make sure the instance in your VPC has the internet connection. A website may disable ping response for a security reason. If you cannot ping google.com, please try to ping another website. If you can successfully ping a website, that means your VPC is created properly.

Please provide screenshots demonstrating how you connected to your instance via SSH, how you ran the ping command, and the resulting output.

```
~ $ aws ec2 describe-instances --instance-ids i-05746087d8cc178ca --query "Reservations[*].Instances[*].PublicIpAddress" --output text
52.87.163.233
~ $
```

```
~ $ ssh -i devenv-key.pem ubuntu@52.87.163.233
The authenticity of host '52.87.163.233 (52.87.163.233)' can't be established.
ED25519 key fingerprint is SHA256:LYKZxvclQ9XY7cUupzTmP33L7nP06BuVPazvjNtB66I.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '52.87.163.233' (ED25519) to the list of known hosts.
Welcome to Ubuntu 20.04.2 LTS (GNU/Linux 5.4.0-1045-aws x86_64)
```

```
* Documentation:  https://help.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:        https://ubuntu.com/advantage
```

System information as of Sat Mar 29 01:38:59 UTC 2025

```
System load:  0.0      Processes:      104
Usage of /:   16.4% of 7.69GB   Users logged in:  0
Memory usage: 23%      IPv4 address for ens5: 10.6.8.13
Swap usage:   0%
```

```
1 update can be applied immediately.
To see these additional updates run: apt list --upgradable
```

```
The list of available updates is more than a week old.
To check for new updates run: sudo apt update
```

```
The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.
```

```
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.
```

```
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.
```

```
ubuntu@ip-10-6-8-13:~$
```

```
ubuntu@ip-10-6-8-13:~$ ping -c 5 google.com
PING google.com (142.251.163.100) 56(84) bytes of data:
64 bytes from wv-in-f100.1e100.net (142.251.163.100): icmp_seq=1 ttl=105 time=1.67 ms
64 bytes from wv-in-f100.1e100.net (142.251.163.100): icmp_seq=2 ttl=105 time=1.66 ms
64 bytes from wv-in-f100.1e100.net (142.251.163.100): icmp_seq=3 ttl=105 time=1.72 ms
64 bytes from wv-in-f100.1e100.net (142.251.163.100): icmp_seq=4 ttl=105 time=1.72 ms
64 bytes from wv-in-f100.1e100.net (142.251.163.100): icmp_seq=5 ttl=105 time=1.66 ms

--- google.com ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4007ms
rtt min/avg/max/mdev = 1.658/1.683/1.718/0.027 ms
ubuntu@ip-10-6-8-13:~$
```

16. Before moving on to the next session, exit out of your EC2 Instance by typing exit. You will need to exit twice if you made yourself root using sudo.

```
ubuntu@ip-10-6-8-13:~$ exit
logout
Connection to 52.87.163.233 closed.
~ $
```

Part 2: NACL (Network Access Control List)



17. Find out the default NACL for your subnet with the following command:

```
aws ec2 describe-network-acls --region us-east-1 --output table --query 'NetworkAcls[*].NetworkAclId' --filters Name=association.subnet-id,Values=your_SubnetId
```

Note - Make sure there is no space between Key=Value ie Values= subnetid. There cannot be any spaces between the equals sign otherwise you will get an error ie

```
[root@localhost mcy611]# aws ec2 describe-network-acls --region us-east-1 --output table --query 'NetworkAcls[*].NetworkAclId' --filters Name=association.subnet-id,Values= subnet-0d61567a0ae6da699
```

Error parsing parameter '--filters': Expected: '=', received: 'EOF' for input:  
subnet-0d61567a0ae6da699

**What are your command and its complete output? What is your network ACL Id?**

```
~ $ aws ec2 describe-network-acls --region us-east-1 --output table --query 'NetworkAcls[*].NetworkAclId' --filters Name=association.subnet-id,Values=subnet-09df9a6645aea6ede
+-----+-----+
| DescribeNetworkAcls |
+-----+-----+
| acl-091a9c3657de4ee08 |
+-----+-----+
```

18. Describe your network ACL with the following command:

```
aws ec2 describe-network-acls --network-acl-ids your_network_acl_id
```

**What are your command and its complete output? Please explain the ACL rules in the output.**

```
~ $ aws ec2 describe-network-acls --network-acl-ids acl-091a9c3657de4ee08
{
  "NetworkAcls": [
    {
      "Associations": [
        {
          "NetworkAclAssociationId": "aclassoc-06cc48ec7eb62debd",
          "NetworkAclId": "acl-091a9c3657de4ee08",
          "SubnetId": "subnet-09df9a6645aea6ede"
        }
      ],
      "Entries": [
        {
          "CidrBlock": "0.0.0.0/0",
          "Egress": true,
          "Protocol": "-1",
          "RuleAction": "allow",
          "RuleNumber": 100
        },
        {
          "CidrBlock": "0.0.0.0/0",
          "Egress": true,
          "Protocol": "-1",
          "RuleAction": "deny",
          "RuleNumber": 32767
        },
        {
          "CidrBlock": "0.0.0.0/0",
          "Egress": false,
          "Protocol": "-1",
          "RuleAction": "allow",
          "RuleNumber": 100
        }
      ]
    }
  ]
}
```

```

    },
    {
      "CidrBlock": "0.0.0.0/0",
      "Egress": false,
      "Protocol": "-1",
      "RuleAction": "deny",
      "RuleNumber": 32767
    }
  ],
  "IsDefault": true,
  "NetworkAclId": "acl-091a9c3657de4ee08",
  "Tags": [],
  "VpcId": "vpc-0cb3e30db617f6175",
  "OwnerId": "881490118823"
}
]
}

```

19. Create an instance in your subnet with the following command:

`aws ec2 run-instances --image-id ami-05c9c0fab0c0bb43e --count 1 --instance-type t3.micro --key-name your_username --security-group-ids your_security_group_id --subnet-id your_SubnetId`

```

~$ aws ec2 run-instances --image-id ami-05c9c0fab0c0bb43e --count 1 --instance-type t3.micro --key-name corrd12-key --security-group-ids sg-0b1cf043fd7ea602b --subnet-id subnet-09df9a6645ae06
ode
{
  "ReservationId": "r-070c9f933b52e1e0c",
  "OwnerId": "881490118823",
  "Groups": [],
  "Instances": [
    {
      "Architecture": "x86_64",
      "BlockDeviceMappings": [],
      "ClientToken": "ebe6e89b-69fc-47f8-9c25-efdb72aac558",
      "EbsOptimized": false,
      "EnaSupport": true,
      "Hypervisor": "xen",
      "NetworkInterfaces": [
        {
          "Attachment": {
            "AttachTime": "2025-03-29T01:49:22+00:00",
            "AttachmentId": "eni-attach-02696a0e487f0e271",
            "DeleteOnTermination": true,
            "DeviceIndex": 0,
            "Status": "attaching",
            "NetworkCardIndex": 0
          },
          "Description": "",
          "Groups": [
            {
              "GroupId": "sg-0b1cf043fd7ea602b",
              "GroupName": "corrd12"
            }
          ],
          "Ipv6Addresses": [],
          "MacAddress": "12:28:60:62:1b:fb",
          "NetworkInterfaceId": "eni-0dd878b14ee194ca3",

```

```
    "NetworkInterfaceId": "eni-0dd878b14ee194ca3",
    "OwnerId": "881490118823",
    "PrivateIpAddress": "10.6.8.5",
    "PrivateIpAddresses": [
      {
        "Primary": true,
        "PrivateIpAddress": "10.6.8.5"
      }
    ],
    "SourceDestCheck": true,
    "Status": "in-use",
    "SubnetId": "subnet-09df9a6645aea6ede",
    "VpcId": "vpc-0cb3e30db617f6175",
    "InterfaceType": "interface",
    "Operator": {
      "Managed": false
    }
  }
],
"RootDeviceName": "/dev/sda1",
"RootDeviceType": "ebs",
"SecurityGroups": [
  {
    "GroupId": "sg-0b1cfa43fd7ea602b",
    "GroupName": "carrd12"
  }
],
"SourceDestCheck": true,
"StateReason": {
  "Code": "pending",
  "Message": "pending"
},
"VirtualizationType": "hvm",
"CpuOptions": {
```

```

    "CpuOptions": {
      "CoreCount": 1,
      "ThreadsPerCore": 2
    },
    "CapacityReservationSpecification": {
      "CapacityReservationPreference": "open"
    },
    "MetadataOptions": {
      "State": "pending",
      "HttpTokens": "optional",
      "HttpPutResponseHopLimit": 1,
      "HttpEndpoint": "enabled",
      "HttpProtocolIpv6": "disabled",
      "InstanceMetadataTags": "disabled"
    },
    "EnclaveOptions": {
      "Enabled": false
    },
    "PrivateDnsNameOptions": {
      "HostnameType": "ip-name",
      "EnableResourceNameDnsARecord": false,
      "EnableResourceNameDnsAAAARecord": false
    },
    "MaintenanceOptions": {
      "AutoRecovery": "default"
    },
    "CurrentInstanceBootMode": "legacy-bios",
    "Operator": {
      "Managed": false
    },
    "InstanceId": "i-07376b78aa6f65f9f",
    "ImageId": "ami-05c9c0fab0c0bb43e",
    "State": {
      "Code": 0,

```

```

    "ImageId": "ami-05c9c0fab0c0bb43e",
    "State": {
      "Code": 0,
      "Name": "pending"
    },
    "PrivateDnsName": "ip-10-6-8-5.ec2.internal",
    "PublicDnsName": "",
    "StateTransitionReason": "",
    "KeyName": "carrd12-key",
    "AmiLaunchIndex": 0,
    "ProductCodes": [],
    "InstanceType": "t3.micro",
    "LaunchTime": "2025-03-29T01:49:22+00:00",
    "Placement": {
      "GroupName": "",
      "Tenancy": "default",
      "AvailabilityZone": "us-east-1c"
    },
    "Monitoring": {
      "State": "disabled"
    },
    "SubnetId": "subnet-09df9a6645aea6ede",
    "VpcId": "vpc-0cb3e30db617f6175",
    "PrivateIpAddress": "10.6.8.5"

```

```

  }
}
~ $ █

```

20. Determine the public IP address of the instance. Which command did you use to find it? What is the public IP address of the instance?

```
~ $ aws ec2 describe-instances --instance-ids i-07376b78aa6f65f9f --query "Reservations[*].Instances[*].PublicIpAddress" --output text
44.211.144.92
```

21. Can you SSH into your instance? Please provide a screenshot showing your SSH login process.

```
~ $ ssh -i devenv-key.pem ubuntu@44.211.144.92
The authenticity of host '44.211.144.92 (44.211.144.92)' can't be established.
ED25519 key fingerprint is SHA256:GNVxUQHNDP/0mwz1H0t9u/Yo4jokVNRax8Aaai+F7gw.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '44.211.144.92' (ED25519) to the list of known hosts.
Welcome to Ubuntu 21.04 (GNU/Linux 5.11.0-1007-aws x86_64)
```

```
* Documentation:  https://help.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:        https://ubuntu.com/advantage
```

System information as of Sat Mar 29 02:00:50 UTC 2025

```
System load:  0.0      Processes:            101
Usage of /:   17.0% of 7.69GB   Users logged in:     0
Memory usage: 20%      IPv4 address for ens5: 10.6.8.5
Swap usage:   0%
```

```
The list of available updates is more than a week old.
To check for new updates run: sudo apt update
```

The programs included with the Ubuntu system are free software;  
the exact distribution terms for each program are described in the  
individual files in /usr/share/doc/\*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by  
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".  
See "man sudo\_root" for details.

```
ubuntu@ip-10-6-8-5:~$
```

22. If you're logged into your instance, you need to exit out of your instance for the next step.

```
ubuntu@ip-10-6-8-5:~$ exit
logout
Connection to 44.211.144.92 closed.
~ $
```

23. Create an entry (a rule) in your network ACL with the following command:

```
aws ec2 create-network-acl-entry --network-acl-id your_network_acl_id --ingress --rule-number 80 --protocol tcp
--port-range From=22,To=22 --cidr-block 0.0.0.0/0 --rule-action deny
```

What is your command? What is the rule?

```
~ $ aws ec2 create-network-acl-entry --network-acl-id acl-091a9c3657de4ee08 --ingress --rule-number 80 --protocol tcp --port
-range From=22,To=22 --cidr-block 0.0.0.0/0 --rule-action deny
```

Deny traffic to port 22

24. Verify if the new rule is added to the default NACL with the following command:

`aws ec2 describe-network-acls --network-acl-ids your_network_acl_id`

**What are your command and its complete output?**

```
~ $ aws ec2 describe-network-acls --network-acl-ids acl-091a9c3657de4ee08
{
  "NetworkAcls": [
    {
      "Associations": [
        {
          "NetworkAclAssociationId": "aclassoc-06cc48ec7eb62debd",
          "NetworkAclId": "acl-091a9c3657de4ee08",
          "SubnetId": "subnet-09df9a6645aea6ede"
        }
      ],
      "Entries": [
        {
          "CidrBlock": "0.0.0.0/0",
          "Egress": true,
          "Protocol": "-1",
          "RuleAction": "allow",
          "RuleNumber": 100
        },
        {
          "CidrBlock": "0.0.0.0/0",
          "Egress": true,
          "Protocol": "-1",
          "RuleAction": "deny",
          "RuleNumber": 32767
        },
        {
          "CidrBlock": "0.0.0.0/0",
          "Egress": false,
          "PortRange": {
            "From": 22,
            "To": 22
          },
          "Protocol": "6",

```

```

        "CidrBlock": "0.0.0.0/0",
        "Egress": false,
        "PortRange": {
            "From": 22,
            "To": 22
        },
        "Protocol": "6",
        "RuleAction": "deny",
        "RuleNumber": 80
    },
    {
        "CidrBlock": "0.0.0.0/0",
        "Egress": false,
        "Protocol": "-1",
        "RuleAction": "allow",
        "RuleNumber": 100
    },
    {
        "CidrBlock": "0.0.0.0/0",
        "Egress": false,
        "Protocol": "-1",
        "RuleAction": "deny",
        "RuleNumber": 32767
    }
],
    "IsDefault": true,
    "NetworkAclId": "acl-091a9c3657de4ee08",
    "Tags": [],
    "VpcId": "vpc-0cb3e30db617f6175",
    "OwnerId": "881490118823"
}
]
}
~ $

```

25. Can you use ssh to log into your instance? Why or why not? Please provide a screenshot showing the SSH login process.

```

~ $ ssh -i devenv-key.pem ubuntu@44.211.144.92
ssh: connect to host 44.211.144.92 port 22: Connection timed out
~ $

```

My NACLs inbound and outbound rules deny SSH traffic on port 22 (TCP) from my IP range.

26. Remove the rule you added in step 15 with the following command:

aws ec2 delete-network-acl-entry --network-acl-id *your\_network\_acl\_id* --ingress --rule-number 80

```

~ $ aws ec2 delete-network-acl-entry --network-acl-id acl-091a9c3657de4ee08 --ingress --rule-number 80
~ $

```

27. Verify if the rule is removed with the following command:

aws ec2 describe-network-acls --network-acl-ids *your\_network\_acl\_id*

What is your command and its complete output?

```
~ $ aws ec2 describe-network-acls --network-acl-ids acl-091a9c3657de4ee08
{
  "NetworkAcls": [
    {
      "Associations": [
        {
          "NetworkAclAssociationId": "aclassoc-06cc48ec7eb62debd",
          "NetworkAclId": "acl-091a9c3657de4ee08",
          "SubnetId": "subnet-09df9a6645aea6ede"
        }
      ],
      "Entries": [
        {
          "CidrBlock": "0.0.0.0/0",
          "Egress": true,
          "Protocol": "-1",
          "RuleAction": "allow",
          "RuleNumber": 100
        },
        {
          "CidrBlock": "0.0.0.0/0",
          "Egress": true,
          "Protocol": "-1",
          "RuleAction": "deny",
          "RuleNumber": 32767
        },
        {
          "CidrBlock": "0.0.0.0/0",
          "Egress": false,
          "Protocol": "-1",
          "RuleAction": "allow",
          "RuleNumber": 100
        }
      ]
    }
  ]
}
```



```

        "RuleAction": "allow",
        "RuleNumber": 100
    },
    {
        "CidrBlock": "0.0.0.0/0",
        "Egress": true,
        "Protocol": "-1",
        "RuleAction": "deny",
        "RuleNumber": 32767
    },
    {
        "CidrBlock": "0.0.0.0/0",
        "Egress": false,
        "Protocol": "-1",
        "RuleAction": "allow",
        "RuleNumber": 100
    },
    {
        "CidrBlock": "0.0.0.0/0",
        "Egress": false,
        "Protocol": "-1",
        "RuleAction": "deny",
        "RuleNumber": 32767
    }
],
    "IsDefault": true,
    "NetworkAclId": "acl-091a9c3657de4ee08",
    "Tags": [],
    "VpcId": "vpc-0cb3e30db617f6175",
    "OwnerId": "881490118823"
}
]
}
~ $

```

28. Can you use ssh to log into your instance? Why or why not? Please provide a screenshot showing the SSH login process.

```

~ $ ssh -i devenv-key.pem ubuntu@44.211.144.92
Welcome to Ubuntu 21.04 (GNU/Linux 5.11.0-1007-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage

System information as of Sat Mar 29 02:15:49 UTC 2025

System load:  0.0               Processes:    101
Usage of /:   17.1% of 7.69GB   Users logged in: 0
Memory usage: 20%              IPv4 address for ens5: 10.6.8.5
Swap usage:   0%

The list of available updates is more than a week old.
To check for new updates run: sudo apt update
Your Ubuntu release is not supported anymore.
For upgrade information, please visit:
http://www.ubuntu.com/releaseendoflife

New release '22.04.5 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Last login: Sat Mar 29 02:00:51 2025 from 3.80.78.79
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

Last login: Sat Mar 29 02:00:51 2025 from 3.80.78.79
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@ip-10-6-8-5:~$ exit
logout
Connection to 44.211.144.92 closed.

```

My NACLs inbound and outbound rule was deleted allowing SSH traffic on port 22 (TCP) from my IP range.

### Part 3: Cleanup

29. Terminate your instances.

**What is your command and its complete output?**

```

~ $ aws ec2 terminate-instances --instance-ids i-05746087d8cc178ca
{
  "TerminatingInstances": [
    {
      "InstanceId": "i-05746087d8cc178ca",
      "CurrentState": {
        "Code": 32,
        "Name": "shutting-down"
      },
      "PreviousState": {
        "Code": 16,
        "Name": "running"
      }
    }
  ]
}
~ $

```

```
~ $ aws ec2 terminate-instances --instance-ids i-07376b78aa6f65f9f
{
  "TerminatingInstances": [
    {
      "InstanceId": "i-07376b78aa6f65f9f",
      "CurrentState": {
        "Code": 32,
        "Name": "shutting-down"
      },
      "PreviousState": {
        "Code": 16,
        "Name": "running"
      }
    }
  ]
}
```

Keep your VPC and subnet for later labs. Write down your VPC id, your subnet id, and your security group id. You will use them for later labs.

#### Submission instruction

After completing this work, submit this word document with your answers to Canvas. In your word document, your answers should be in **bold** print. At the end of your word document, you should write a few sentences that explain the goals of this lab and a paragraph or two that summarizes (high level) the steps one needs to take in order to complete those goals.

I began by setting up a Virtual Private Cloud (VPC) and a subnet inside of it in order to configure my AWS setup.

After making sure all the required setups were in place, I started EC2 instances in the subnet. To provide remote access to the instances, I put in place a security group that permitted incoming SSH connections on port 22. A

Network Access Control List (NACL) was also configured by me to control traffic at the subnet level. I made sure that the security group and NACL rules were correctly implemented and functioning as planned by trying to connect via SSH after establishing these components.